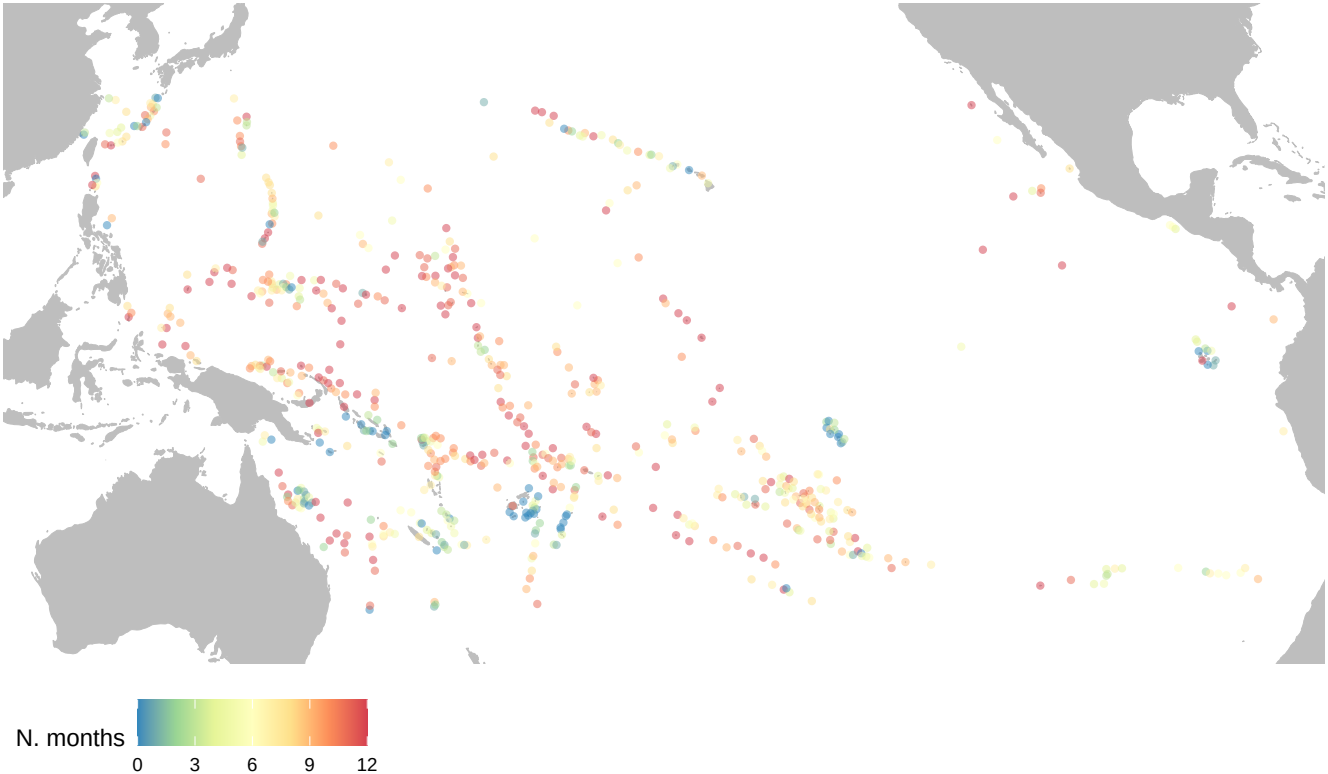
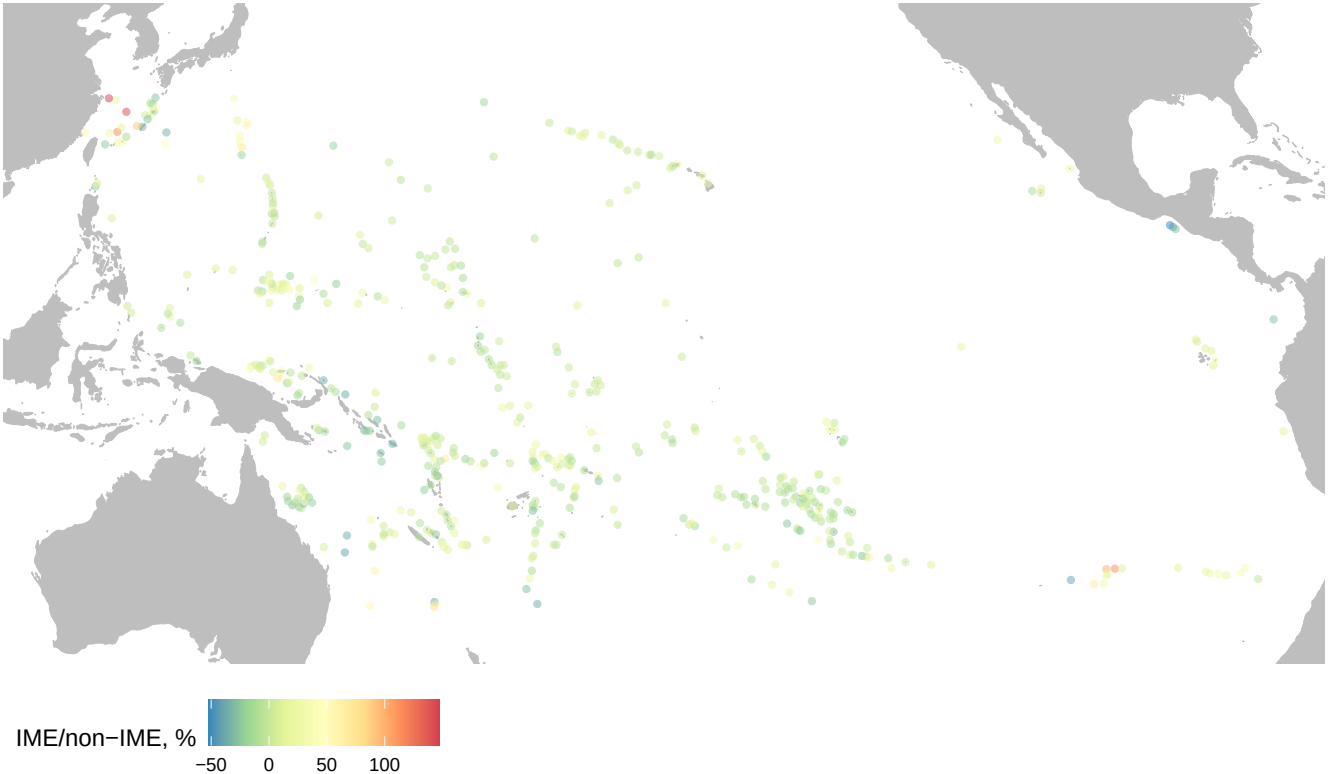
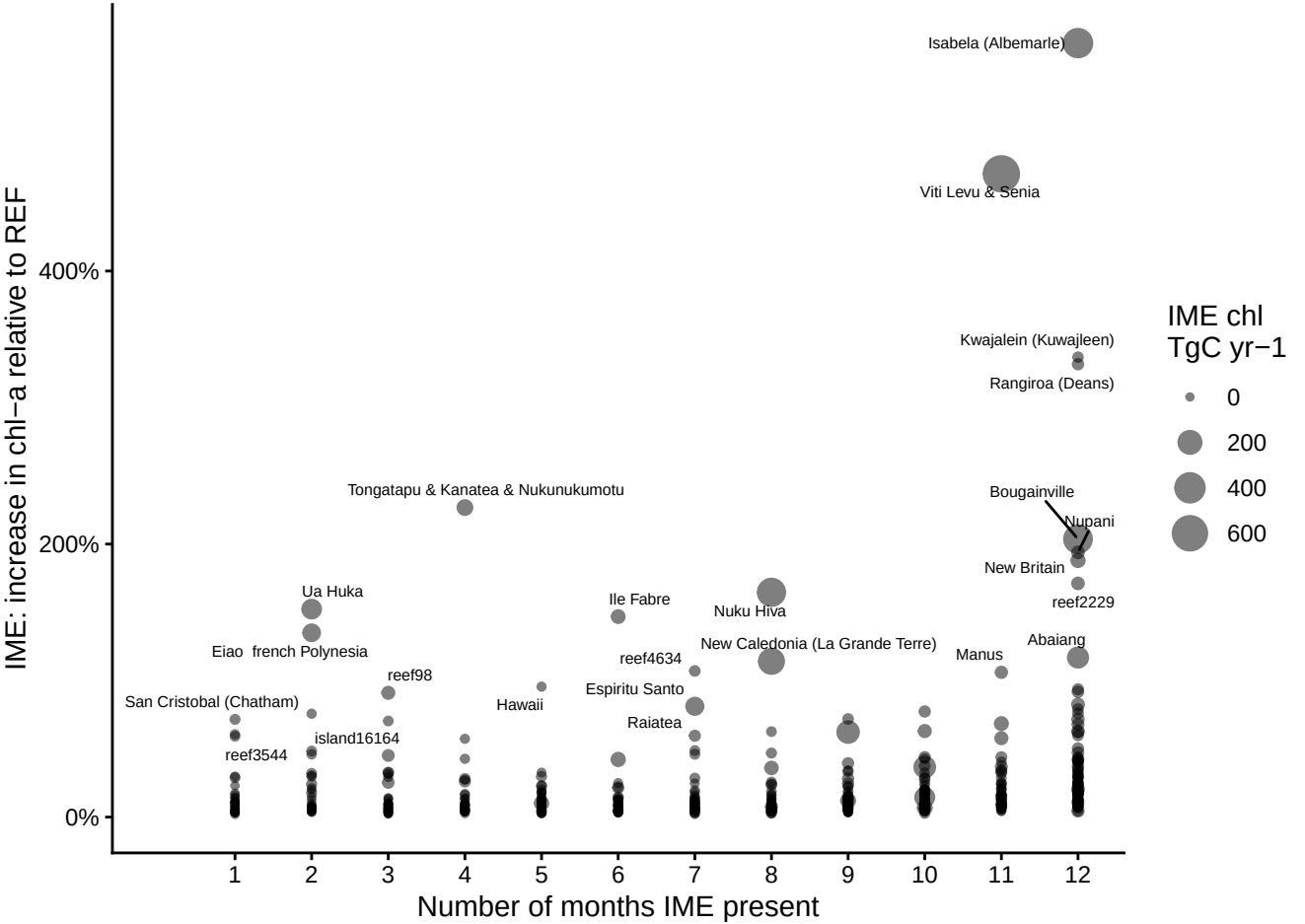
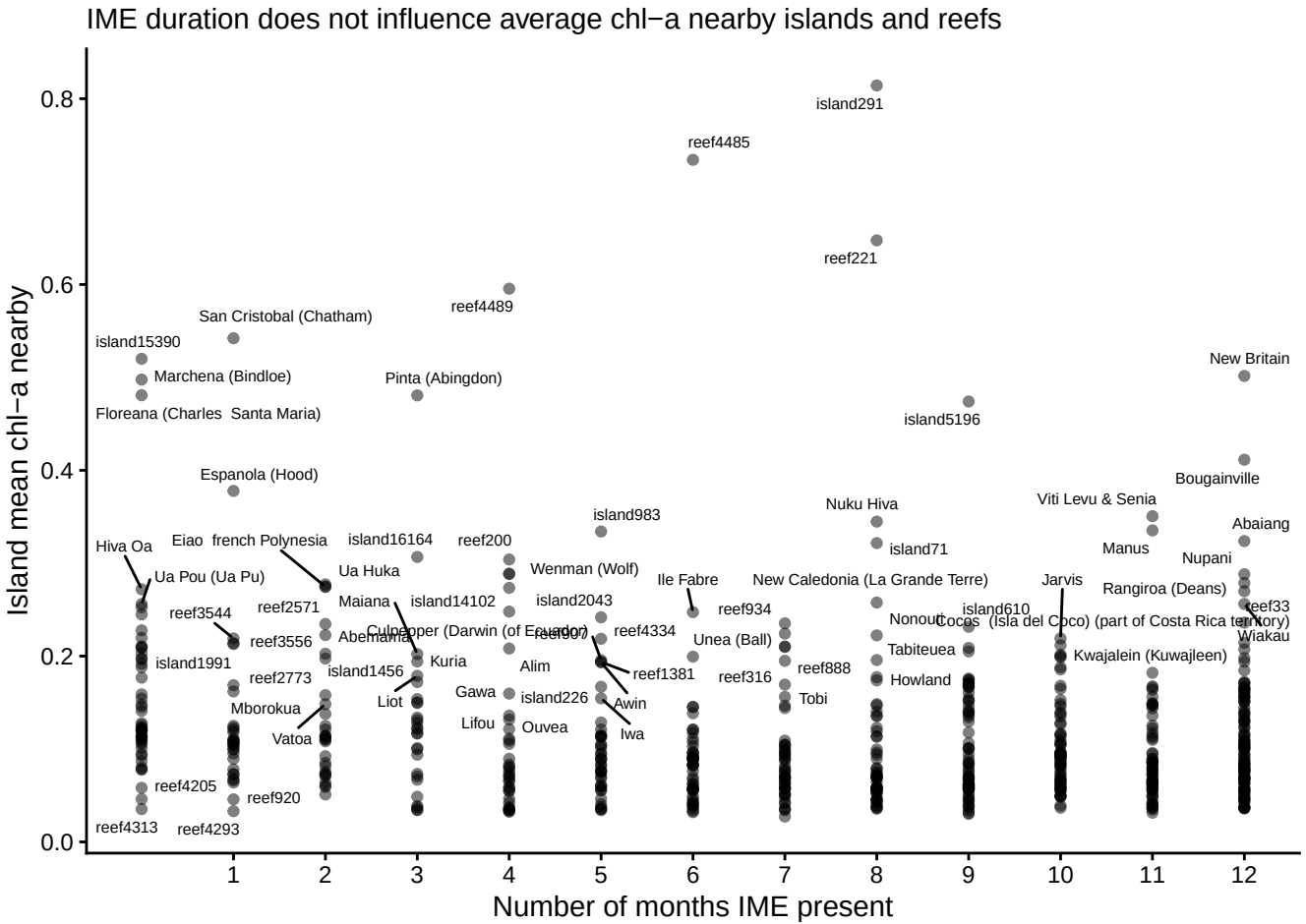
Seasonality in IME duration

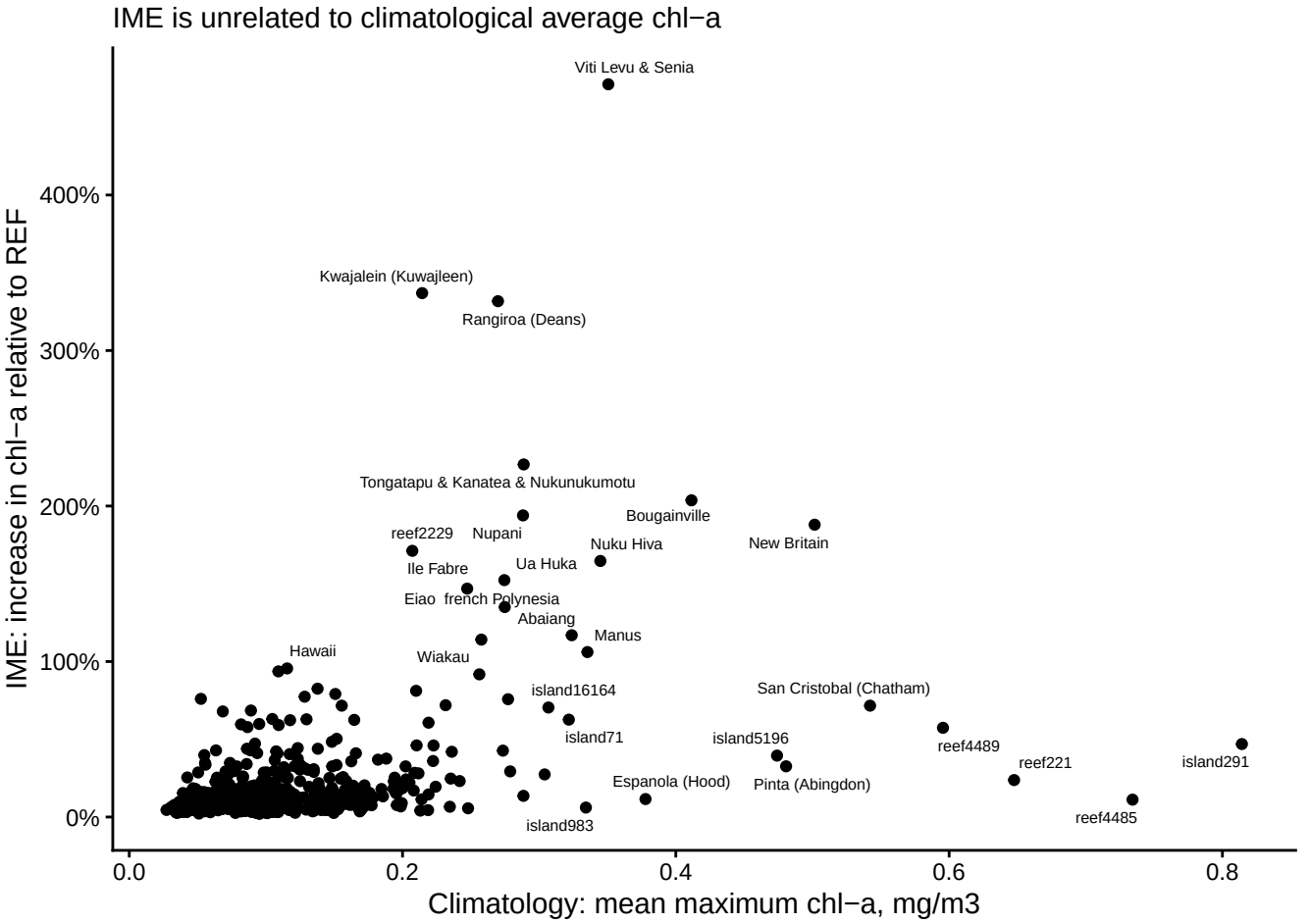


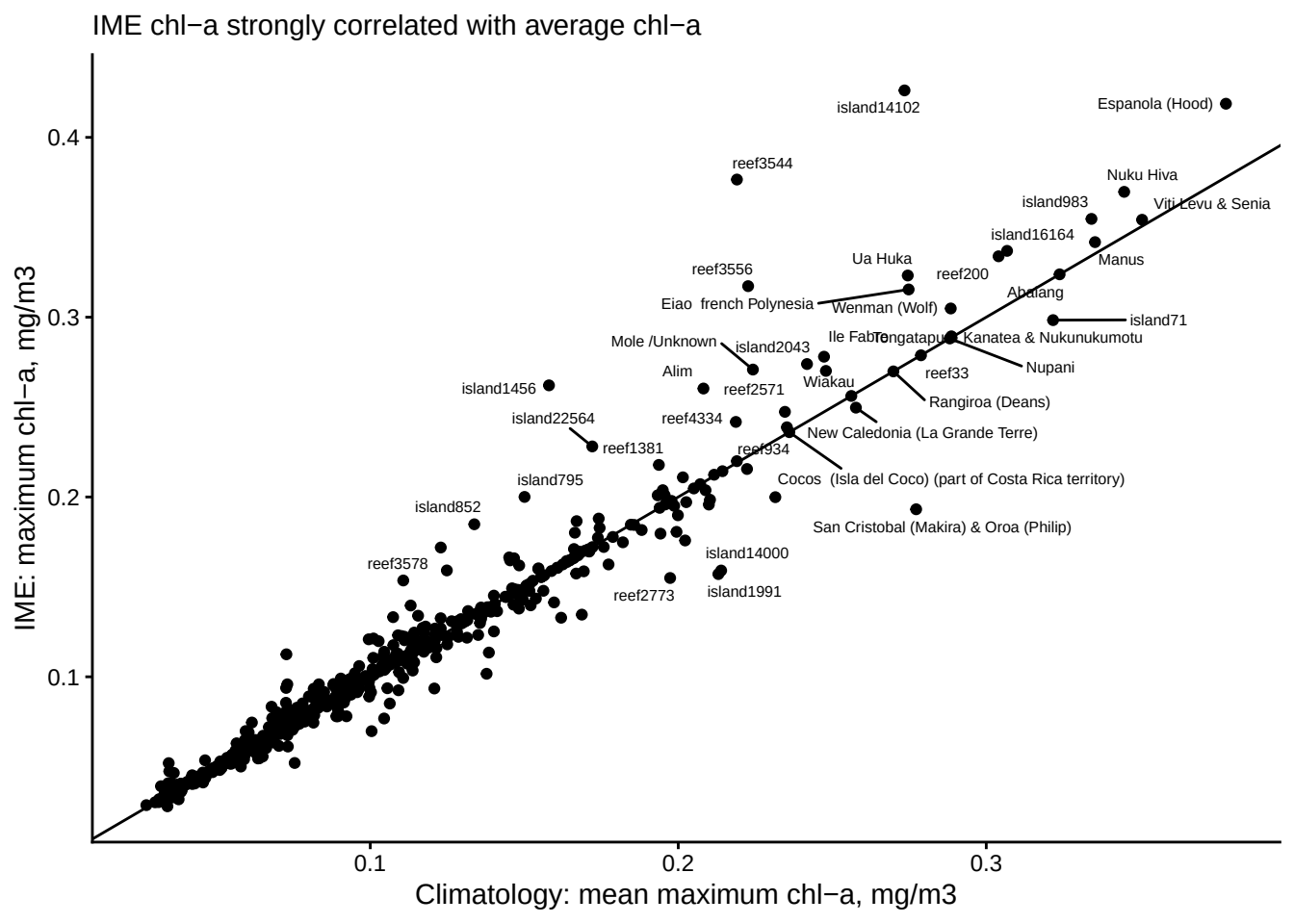
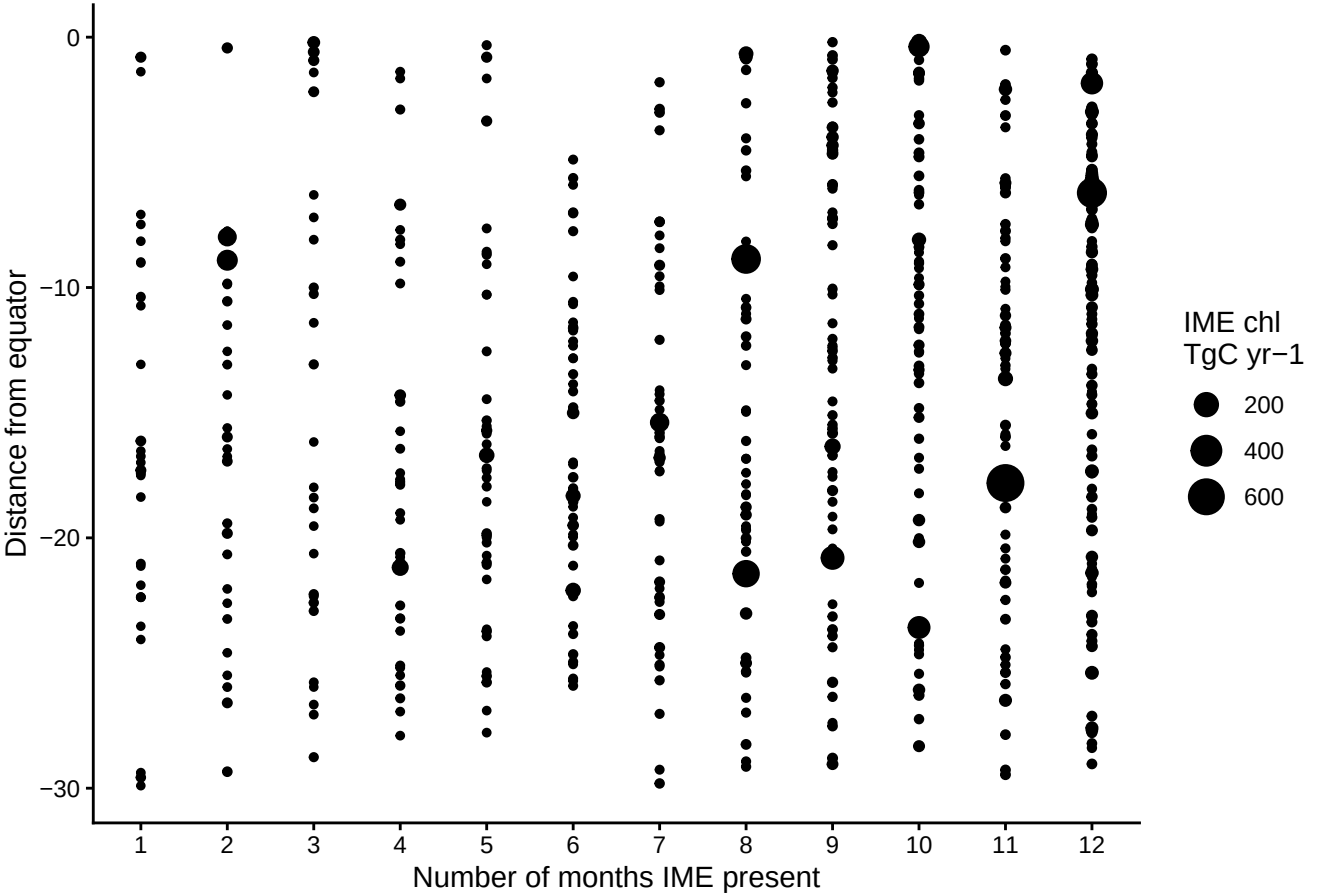
Non−IME periods can have higher chl−a than IME




IME: increase in chl−a relative to REF
400%
200%
0%
Isabela (Albemarle)
Viti Levu & Senia
Kwajalein (Kuwajleen)
Rangiroa (Deans)
Tongatapu & Kanatea & Nukunukumotu
Bougainville
Nupani
New Britain
reef2229
Ua Huka
Ile Fabre
Nuku Hiva
Eiao
french Polynesia
New Caledonia (La Grande Terre)
Abaiang
Manus
reef4634
reef98
Espiritu Santo
San Cristobal (Chatham)
Hawaii
Raiatea
island16164
reef3544
1
2
3
4
5
6
7
8
9
10
11
12
Number of months IME present
IME chl
TgC yr−1
0
200
400
600


IME duration does not influence average chl−a nearby islands and reefs
Island mean chl−a nearby
Number of months IME present
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
6
7
8
9
10
11
12
island291
reef4485
reef221
reef4489
San Cristobal (Chatham)
island15390
Marchena (Bindloe)
Floreana (Charles Santa Maria)
Pinta (Abingdon)
island5196
New Britain
Espanola (Hood)
Bougainville
Nuku Hiva
Viti Levu & Senia
island983
Abaiang
Hiva Oa
Eiao
french Polynesia
island16164
reef200
island71
Manus
Nupani
Ua Pou (Ua Pu)
Ua Huka
Wenman (Wolf)
Ile Fabre
New Caledonia (La Grande Terre)
Jarvis
Rangiroa (Deans)
reef3544
reef2571
Maiana
island14102
island2043
reef934
island610
reef33
reef3556
Abemama
Culpepper (Darwin (of Ecuador))
reef997
reef4334
Unea (Ball)
Nonouti
Cocos (Isla del Coco) (part of Costa Rica territory)
Wiakau
island1991
reef2773
Kuria
Alim
reef1381
reef316
reef888
Tabiteuea
Kwajalein (Kuwajleen)
island1456
Howland
Mborokua
Liot
Gawa
island226
Awin
Tobi
Lifou
Ouvea
Iwa
Vatoa
reef4205
reef920
reef4313
reef4293


IME is unrelated to climatological average chl−a
IME: increase in chl−a relative to REF
400%
300%
200%
100%
0%
0.0
0.2
0.4
0.6
0.8
Climatology: mean maximum chl−a, mg/m3
Viti Levu & Senia
Kwajalein (Kuwajleen)
Rangiroa (Deans)
Tongatapu & Kanatea & Nukunukumotu
Bougainville
reef2229
Nupani
Nuku Hiva
New Britain
Ile Fabre
Ua Huka
Eiao
french Polynesia
Abaiang
Manus
Hawaii
Wiakau
island16164
San Cristobal (Chatham)
island71
island5196
reef4489
reef221
island291
Espanola (Hood)
Pinta (Abingdon)
reef4485
island983


IME chl−a strongly correlated with average chl−a
IME: maximum chl−a, mg/m3
Climatology: mean maximum chl−a, mg/m3
0.1
0.2
0.3
0.4
island14102
Espanola (Hood)
reef3544
Nuku Hiva
island983
Viti Levu & Senia
island16164
Manus
Ua Huka
reef200
reef3556
Abaiang
Eiao
french Polynesia
Wenman (Wolf)
island71
Mole /Unknown
island2043
Ile Fabre
Tengatapu
Kanatea & Nukunukumotu
Alim
Nupani
reef33
island1456
reef2571
Wiakau
Rangiroa (Deans)
island22564
reef4334
New Caledonia (La Grande Terre)
reef1381
reef934
Cocos (Isla del Coco) (part of Costa Rica territory)
island795
island852
San Cristobal (Makira) & Oroa (Philip)
island14000
reef3578
reef2773
island1991

Temporal IME variability occurs across Pacific Ocean